Interacting effects of droplet number and ice formation processes on mixed-phase cold-air outbreak clouds

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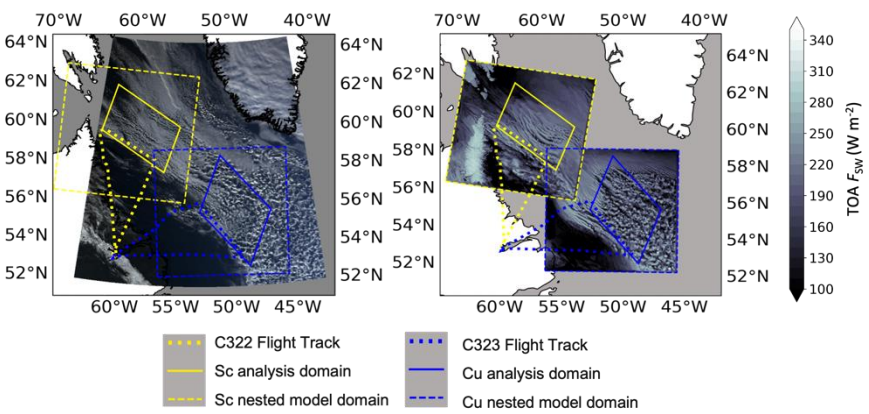
Large uncertainties exist in estimating how high-latitude mixed-phase clouds respond to the warming of climate.[1,2]

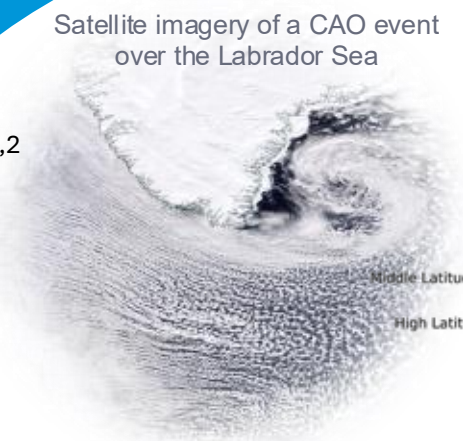
Cold-air outbreak events are natural laboratories to study the effects of aerosol-cloud interactions and ice formation processes on the radiative properties of mixed-phase clouds.[3]



Top figure: (left) satellite RGB imagery of the selected CAO case; (right) modelled TOA shortwave flux

Bottom figure: paired projections for all six perturbed parameters in the (left) stratocumulus region, and the (right) cumulus region.

Satellite imagery of a CAO event over the Labrador Sea



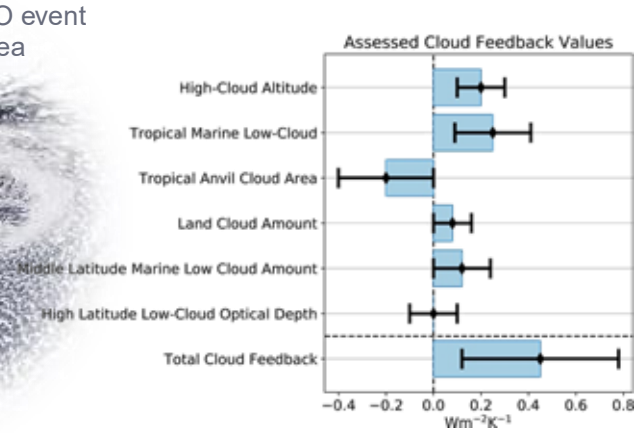


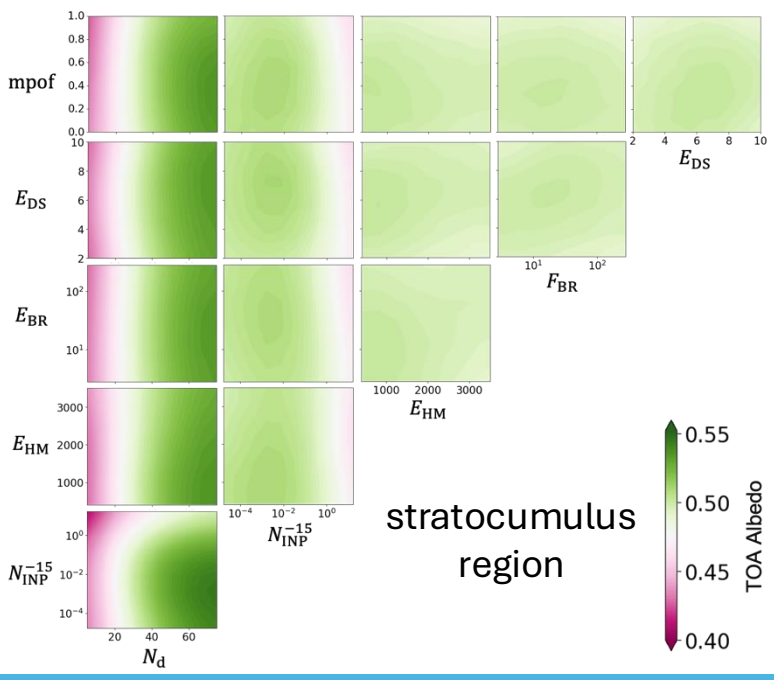
Figure from Sherwood *et al.* (2020)

Previous modelling works on mixed-phase CAO clouds used **one-at-a-time (OAT) test**, which **limits** the exploration of the full model response surface.

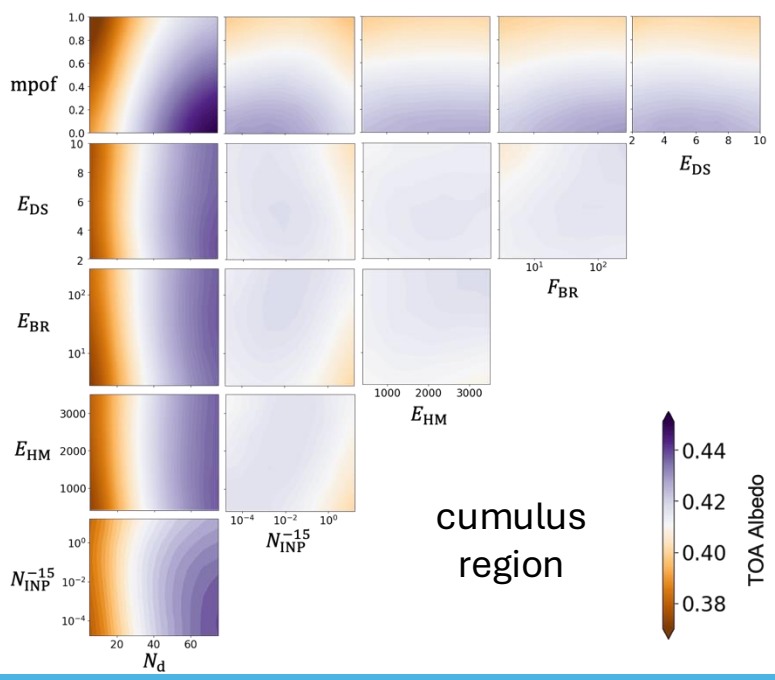
Perturbed parameter ensemble (PPE) simulations using high-resolution model (UK Met Office Unified Model) coupled to a double-moment cloud microphysics scheme, **CASIM** (Cloud AeroSol Interacting Microphysics)

6 parameters: N_d, N_{INP}, E_{HM}, E_{BR}, E_{DS}, and *mpof*

Sensitivities of TOA (top-of-the-atmosphere) albedo to N_d depend on the value of N_{INP}, vice versa



Interactions (non-linear joint effects) between most of the paired parameters on TOA albedo



About me

Final-year PhD researcher in the Aerosols, Clouds & Climate (ACC) Group at the University of Leeds

I love guinea pigs, and here are my two boys, Kapi and Bara.

Refs: 1. **Sherwood, S. C. *et al.*:** An Assessment of Earth's Climate Sensitivity Using Multiple Lines of Evidence, Reviews of Geophysics,2020; 2. **Ceppi, P. *et al.*:** Cloud feedback mechanisms and their representation in global climate models, WIREs Climate Change, 2017; 3. ***Murray, B. J., Carslaw, K. S., and Field, P. R.***: Opinion: Cloud-phase climate feedback and the importance of ice-nucleating particles. Atmospheric Chemistry and Physics, 2021.